

Getset Kiddies
GPS Tracker & Monitoring
with Text-to-Speech Finder Locator

A Capstone Project

Presented to the Faculty of the
College of Information Technology Education
PHINMA University of Iloilo

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Bachelor of Science in Information Technology

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Chapter 1

Introduction to the Study

The goal of this capstone project is to design and implement *Getset Kiddies*, a GPS Tracker and Monitoring System with a Text-to-Speech Finder Locator. This system aims to enhance child safety by providing real-time tracking, location monitoring, and voice-guided alerts. By integrating GPS technology and text-to-speech features, the system will help parents or guardians locate their children easily, by proving quick response during emergencies.

The project focuses on developing a user-friendly interface for parents or guardians to monitor their children's location efficiently. It will include features such as real-time GPS tracking, geofencing alerts, and voice output for easy navigation. The system will be designed for accessibility, offering mobile app compatibility for seamless use in daily activities.

This project aims to provide a reliable and practical solution for improving child safety and reducing stress for caregivers, making sure that the children are always within safe boundaries.

Background of the Study

Child safety has always been a priority for parents, especially in today's busy world. Traditional child monitoring methods, such as close monitoring or simple location sharing, sometimes fail when it comes to providing updated information and immediate help. Examples of modern technologies that

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provide new solutions to these issues are text-to-speech systems and GPS tracking.

To increase safety, a GPS tracker and monitoring system integrates modern technology.

Key features:

1. GPS Technology offers exact, real-time tracking, giving parents to easily locate their child at any time.
2. Text-to-Speech (TTS) by putting location updates into voice commands, text-to-speech (TTS) makes it easier for parents to focus on what really matters and keeps them from distractions.
3. Geofencing children stay inside online safety borders using geofencing, which notifies parents when they leave or enter a specified safe area again.
4. Mobile Application using a user-friendly application provides a smooth and secure experience for the users.
5. Cloud Data Integration for greater security, cloud data integration makes sure that all location data is safely stored and always available. By bringing together these technologies "GetSet Kiddies" provides a complete and useful strategy for protecting kids and providing parents with the latest and useful information.

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Statement of the Problem

One of the most important but challenging responsibilities for parents and guardians is protecting their children. Traditional monitoring techniques can sometimes be unavailable, unsuccessful, and without real-time updates, which can lead to stress and a delayed emergency response.

Specific problems:

Limited Real-Time Updates monitoring may be limited due to outdated technology or unstable internet connection.

Lack of Effective Communication parents usually struggle to find their child when it comes to crowded and unfamiliar places.

Inconvenience of Manual Tracking manual searching can be difficult and time-consuming for a Parent when it comes to public places and can cause a delay in urgent situations.

Difficulty in Emergency Response responding to emergency situations can sometimes be challenging without any specific location or information.

The purpose of this project is to help parents locate, monitor, provide real time updates, and stay informed about their child's activity while ensuring their safety.

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Objectives of the Study

This study aims to create a real-time GPS tracking system for accurate location monitoring, paired with a user-friendly application for easy navigation and operation. It will integrate geofencing technology to define and monitor designated areas, ensuring timely alerts. Data will be stored in secured cloud-based storage to guarantee privacy and protection. The system is designed to be easy to implement in daily life and provide convenient access for all users

The objectives:

1. To develop a real-time GPS tracking system that easily monitors and tracks the location in real time.
2. To develop a user-friendly mobile application allowing parents to conveniently monitor their child's location in real time.
3. To integrate geofencing technology to notify parents when their child leaves the designated areas.
4. To secure cloud-based storage allowing parents to review their child's activity while protecting data privacy.
5. To make sure the system is cost effective and easy to implement in daily life offering practical and accessible solutions to the users.

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Significance of the Study

The significance of this study is to offer a practical solution to the parents struggling to monitor their child's activity. This system lowers the chance of a child getting lost and makes sure that the parents get notified in emergency situations.

The key benefits of this study include:

1. Improved Child Safety provides real-time updates and alerts, reducing risks in emergencies.
2. Convenience providing user friendly features makes monitoring easier for parents.
3. Stress Reduction secure child's safety and knowing their child's whereabouts make parents feel more at ease.
4. Accessibility this app is designed to be used easily making sure that parents can quickly access and understand the features making it convenient and practical for everyday use.

Scope and Limitation of the Study

The purpose of this study is to develop "Getset Kiddies" is designed to monitor and track users. Our system offers real time location tracking, geofencing, and text to speech features that will help the people with physical disabilities providing convenience to the users. Our group targets to offer practical solutions about child safety. But there are some limitations. Our system relies on strong signal and internet

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connection, and it can affect the accuracy of the app. Also the app will not address the specific needs or customization of the user.

Scope:

Real-time location tracking and updates using GPS.

Development of a mobile application for monitoring and alerts.

Integration of geofencing and text-to-speech technologies.

Cloud-based storage for location and activity data.

Limitations:

Hardware Dependence the system requires GPS-enabled devices for functionality.

Internet Connectivity the system relies on internet access for real-time updates and cloud storage.

Geographic Constraints GPS accuracy may vary depending on location and environmental factors.

Limited Customization the system may not address specific needs beyond child monitoring.

Definition of Terms

1. **GPS Tracking:** Technology that determines the exact location of a device using satellite signals.

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2. Text-to-Speech (TTS): Converts text information into spoken voice output.
3. Geofencing: A virtual boundary set around a location, triggering alerts when the boundary is crossed.
4. Cloud Storage: Remote data storage that allows users to access data from multiple devices.
5. Real-Time Monitoring: Tracking and updating location information as it happens.
6. Mobile Application: Software designed for smartphones to access and manage system features.
7. User-Friendly Interface: An easy-to-navigate design that ensures the system is simple to use.
8. Emergency Response: Actions taken to address urgent situations, supported by real-time location updates.
9. Accessibility: The ability to use the system conveniently from different devices or locations.

Chapter 2

Review of Related Literature

An overview of the studies and research that was used when creating Getset Kiddies: GPS Tracker & Monitoring with Text-to-Speech Finder Locator is given in this chapter. This highlights ideas from Speech Synthesis and Recognition by Wendy Holmes and Text-to-Speech Synthesis by Paul Taylor as it

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analyzes important technologies including GPS tracking, child monitoring systems, and text-to-speech technology.

In today's time, child monitoring systems often use GPS tracking as a main feature. GPS allows accurate and real-time location tracking so that parents or guardians can know where their child is at any moment. This technology has become an important part of child safety tools.

Text-to-speech (TTS) technology has also improved greatly over the years. Earlier systems used rule-based or concatenative methods, which often sounded robotic and less natural. Taylor's (2009) book *Text-to-Speech Synthesis*, published by Cambridge University Press, explains the early development of TTS. It discusses linguistic analysis, acoustic modeling, and statistical methods such as hidden Markov models. These early ideas form the foundation for understanding how TTS systems have developed over time.

Building on these foundations, Tan's (2023) *Neural Text-to-Speech Synthesis*, published by Springer, introduces modern TTS systems based on deep learning. These systems are designed to create speech that sounds more natural, expressive, and human-like. Tan also explains how these systems improve efficiency, handle limited data, and perform well in different conditions.

TTS technology is also used in healthcare. Fisher, Metcalf, Pruthi, and Pappas (2020), in their book *Voice Technology in Healthcare: Leveraging Voice to Enhance Patient and Provider Experiences*, describe how voice assistants, wearable devices, and smart speakers help improve communication between patients and healthcare workers. Their research shows that

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well-designed voice systems can make patient interactions easier, save time, and help in emergencies.

Aside from technical progress, ethical and social factors are also important in using TTS responsibly. *The Parents' Survival Guide to Online Safety* by KidGuard Publishing (n.d.) highlights the need for clear and safe digital communication to protect children online. Similarly, *The Parents' Guide to Preventing Child Abduction* by the KidGuard Team (n.d.) focuses on the importance of giving parents timely and easy-to-understand information during emergencies. Although these books do not focus directly on TTS, they give helpful insights into designing systems that keep children safe and informed.

Reference

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Chapter 3

Methodology

This chapter explains the research and development process of the Getset Kiddies GPS Tracker and Monitoring System with Text-to-Speech Finder Locator. The methodology provides a structured approach to designing, developing, and evaluating the system, ensuring it meets the needs of parents or guardians while improving child safety.

Research and Development (R&D) Stage

The R&D stage is a vital phase in creating the Getset Kiddies system. During this stage, development was guided by input from potential users, including parents, guardians, and child safety experts. Their insights helped shape the design, functionality, and usability of the system to ensure it is practical and effective in real-world scenarios.

Initially, research began with surveys, interviews, and observations to identify the challenges parents face when monitoring their children. The feedback highlighted the need for real-time location tracking, geofencing alerts, and voice-guided notifications, forming the foundation of the system.

The system was then designed and developed step by step, incorporating feedback at each stage. Modern technologies such

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as GPS for location tracking, Text-to-Speech for alerts, mobile applications for user interface, and cloud storage for secure data management were integrated to ensure a user-friendly and reliable system.

Regular testing and evaluations were conducted throughout the development phase to refine the system, making sure it is intuitive, accurate, and responsive in emergencies

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) provided a structured approach to developing Getset Kiddies systematically and efficiently. The Agile methodology was chosen because it supports iterative development, continuous feedback, collaboration, and flexibility, which are critical for creating a user-centered system.

1. Iterative Development - The system was developed in small increments, allowing gradual improvements.
2. Continuous Feedback - Parents and experts provided ongoing feedback, helping refine the system.
3. Collaboration - Close communication between developers, users, and advisors ensured alignment with real needs.
4. Flexibility - Changes in requirements could be accommodated at any stage of development.
5. Planning and Requirement Analysis - In this phase, the team gathered detailed requirements from parents, guardians, and child safety experts. Surveys and interviews were conducted

to identify key problems, such as delayed emergency response, difficulty in crowded places, and lack of real-time updates. This stage also defined the scope, goals, and timeline of the project.

6. System Design - The design phase focused on creating both the user interface and the system architecture. Feedback from respondents guided the design to ensure it is user-friendly and accessible. GPS, geofencing, text-to-speech technology, and cloud storage were incorporated to make the system reliable and secure.
7. Development - During development, the actual coding and system integration took place. The mobile app, GPS tracking modules, and text-to-speech alerts were implemented. Early prototypes were tested by users, and feedback was incorporated to improve accuracy, responsiveness, and usability.
8. Testing - Testing ensured that the system worked correctly and efficiently. Functional testing verified that features such as real-time tracking, alerts, and text-to-speech notifications were operational. Performance testing checked that the system could handle multiple users and devices simultaneously. User Acceptance Testing (UAT) involved parents testing the system in simulated real-world scenarios, providing valuable feedback for final refinements.
9. Deployment - After successful testing, the system was deployed for actual use. Parents were given guidance on installing and using the app. The deployment phase ensured the system was fully operational, accessible, and easy to use.
10. Maintenance and Support - Continuous maintenance ensures the system stays up-to-date and functional. Feedback from

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users helps identify areas for future updates, bug fixes, and feature enhancements. Regular monitoring ensures the system continues to provide reliable child safety services.

By adopting this research design, the study ensured a structured and step by step approach where both user requirements and technological considerations were addressed. This allowed the final output to be not only functional and reliable but also user-friendly and accessible for its intended audience.

Data Collection Techniques

To understand the needs of parents and design a functional system, the following methods were used:

1. Surveys - surveys collected quantitative data from parents to identify common safety concerns, expectations, and preferred features.
2. Interviews - in-depth interviews were conducted with selected parents and child safety experts to understand challenges in monitoring children and how technology can help.
3. Focus Groups - small group discussions were organized to gather multiple perspectives on usability, features, and potential improvements.
4. Observations - observing daily routines of children and parents helped identify real-world challenges in child monitoring.

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5. Document Analysis - existing research, reports, and case studies on GPS tracking, child monitoring, and text-to-speech systems were analyzed to identify best practices and technological solutions.

Justification for Selecting the Agile Methodology

The Agile methodology was chosen over traditional models like Waterfall because of its

Criteria	Agile	Waterfall
Flexibility	High - accommodates changes at any stage	Low - changes difficult after design
User Involvement	Continuous feedback and collaboration	Limited to initial and final stages
Development Speed	Faster delivery through incremental sprints	Slower due to sequential phases
Risk Management	Early detection and resolution of issues	Risks may surface late in development

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Criteria	Agile	Waterfall
Flexibility	High - accommodates changes at any stage	Low - changes difficult after design
User Involvement	Continuous feedback and collaboration	Limited to initial and final stages
Development Speed	Faster delivery through incremental sprints	Slower due to sequential phases
Project Suitability	Ideal for evolving requirements and user feedback	Best for well-defined, static projects

Agile allowed the team to respond quickly to user feedback, adjust features, and ensure the system effectively meets the real needs of parents and guardians.

System Architecture

The Getset Kiddies GPS Tracker and Monitoring System consists of three main layers: Device Layer, Server Layer (MySQL Database), and Application Layer. The chosen programming languages optimize each layer for performance and usability:

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C++ for Arduino, Python for backend, and JavaScript for mobile/web functionality.

1. Device Layer (C++ - Arduino)

- GPS Module - captures real-time latitude and longitude.
- GSM/Cellular Module - sends location data to the server.
- Arduino Microcontroller - processes GPS and sensor data.
- Text-to-Speech Module - converts alerts/messages to voice for children.
- Programming C++ is used for Arduino programming to handle
- Reading GPS coordinates
- Communicating with the GSM module
- Triggering TTS output for alerts
- Sending SOS signals

2. Server Layer (NodeJS + MySQL Database)

- Backend Application - python handles server-side logic, including
- Receiving data from Arduino devices via API
- Processing location and alert data
- Checking geofence boundaries
- Storing and querying data in MySQL
- MySQL Database - stores structured data such as Child profiles and device IDs
- Real-time and historical GPS locations
- Geofence boundaries and alert logs

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3. Application Layer (JavaScript - Mobile/Web Functionality)

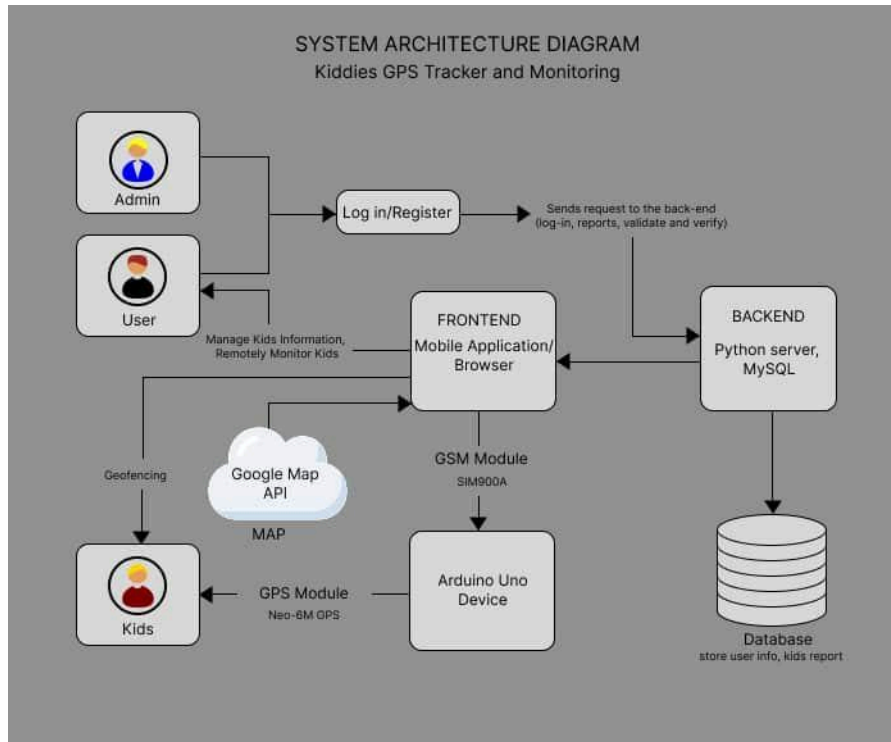
Mobile Application / Web Interface provides parents with:

- Real-time location tracking
- Geofence management
- Alerts and notifications
- Notification System: Push notifications, SMS, or voice alerts delivered to users.
- Programming JavaScript is used for Frontend interface (mobile app frameworks like NodeJS or web dashboard EJS)
- Rendering dynamic maps and real-time location data
- Handling user interactions (setting geofence, viewing history, receiving alerts)

4. Communication Flow

- Arduino (C++) → Python Backend - device sends GPS and alert data via GSM/Internet to Python server.
- Python Backend → MySQL Database - backend processes data and stores it in MySQL.
- Python Backend → JavaScript Frontend - backend sends processed data (location updates, alerts) to mobile/web app.
- JavaScript Frontend → Arduino - parents can send commands through the app (request location, trigger alerts), which are routed via backend to the Arduino device.

System Architecture Diagram



Research Components and Modules

This part explains the main components and modules that make up the *GetSet Kiddies: GPS Tracker and Monitoring System with Text-to-Speech Finder Locator*. Each module has an important role in helping parents track, monitor, and locate their child safely and easily. These parts work together to make sure the system gives real-time updates and quick alerts when needed.

1. GPS Tracking Module

The GPS Tracking Module is the part of the system that finds and sends the child's real-time location. It uses a GPS device attached to the child's tracker to get the exact position and sends it to the server. Parents can then see their child's

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live location on the mobile app. This module makes sure that parents always know where their child is.

2. Text-to-Speech (TTS) Finder Locator Module

The Text-to-Speech Finder Locator Module helps parents find their child by using voice messages. When the parent activates the "Find My Child" feature, the child's device plays a voice message such as "I am here!" This makes it easier for parents to locate their child in crowded places or during emergencies. It adds an extra layer of safety and communication.

3. Geofencing Module

The Geofencing Module allows parents to set safe areas, such as home, school, or playgrounds. When the child leaves or enters these areas, the system automatically sends a message or alert to the parent's phone. This helps parents know right away if their child goes beyond safe zones.

4. Management Module

The Management Module is the main control center of the system. Parents use it to manage their account, add child profiles, connect devices, and set up geofences. It also shows alerts, battery status, and system settings in one easy place. This module keeps everything organized and simple for parents to use.

5. Database Management Module

The Database Management Module stores all important data such as child profiles, locations, and alerts. It uses a MySQL database to safely keep records and make sure that data is accurate and secure. This allows parents to review their child's location history and updates anytime they want.

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6. Communication Module

The Communication Module connects all parts of the system. It makes sure that the GPS device, server, and mobile app can send and receive messages smoothly. It handles sending GPS data, alerts, and commands between the child's device and the parent's application, keeping the system running in real time.

7. Mobile and Web Application Module

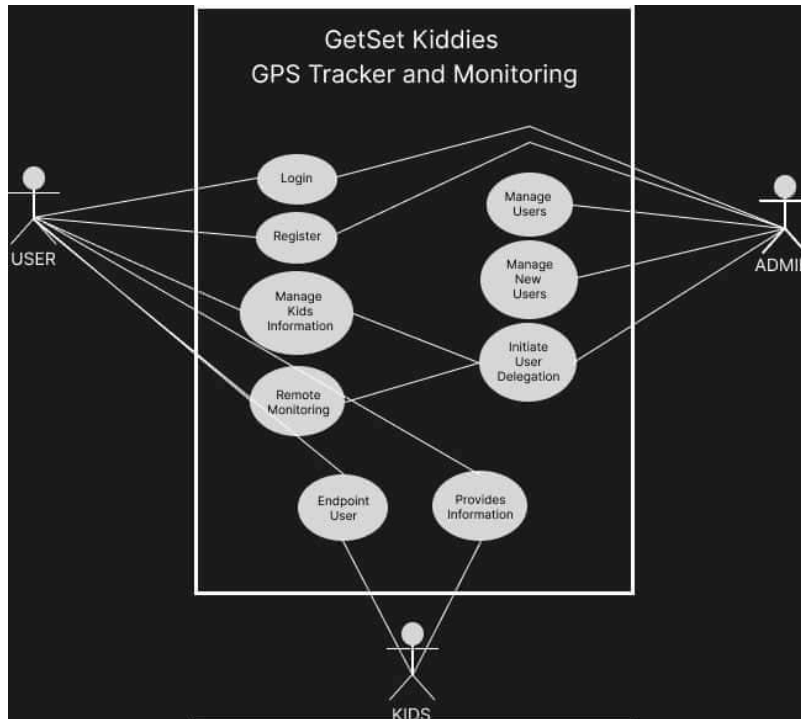
The Mobile and Web Application Module is the part that parents use directly. It shows maps, alerts, and child information in a simple and easy-to-use design. Parents can check their child's location, set safe zones, or use the TTS feature through this app. It is made using JavaScript and works on both phones and computers.

8. Notification and Alert Module

The Notification and Alert Module sends real-time messages or alerts to parents when something important happens, like when the child leaves a safe area or sends an SOS signal. These alerts can appear as text messages or app notifications, helping parents respond quickly in any situation.

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Use Case Diagram



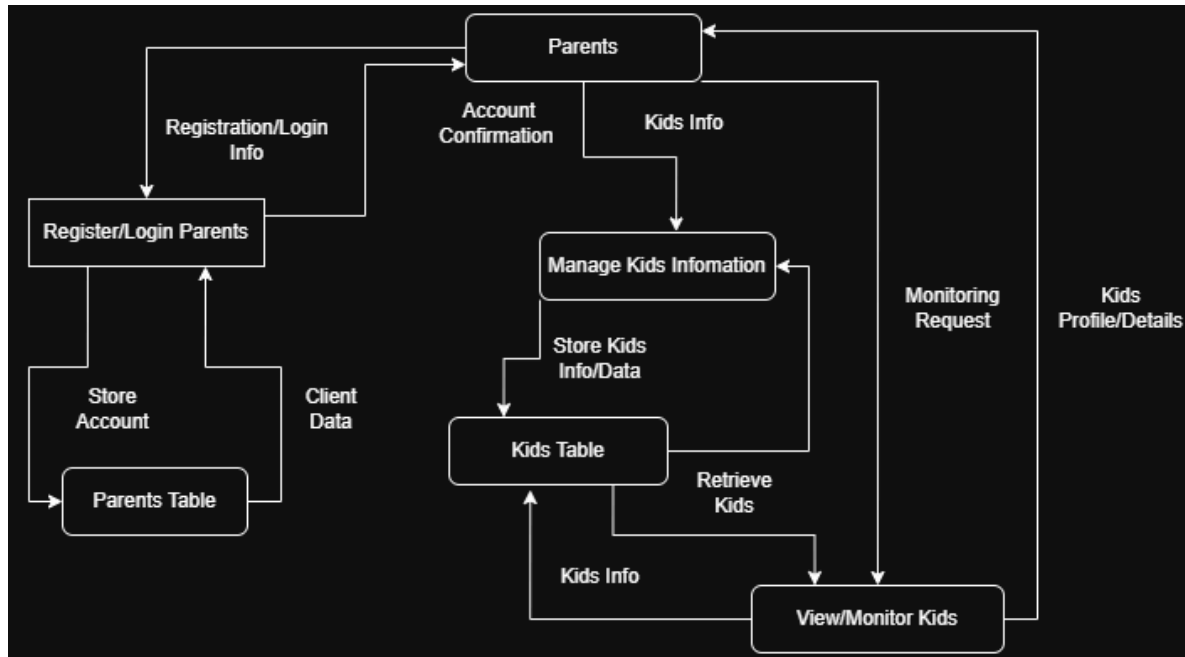
The GetSet Kiddies GPS Tracker and Monitoring System helps parents know where their kids are. A small device with GPS and GSM is carried by the child. It sends the child's location to the parent's phone or computer. Parents can see the location on Google Maps, set safe areas (geofences), and get SMS alerts if the child leaves the area. This system helps parents watch over their kids and keep them safe.

Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) of *Getset Kiddies: GPS Tracker & Monitoring with Text-to-Speech Finder Locator* illustrates the logical flow of data between the user (Parents), the system's internal processes, and the database. It provides a clear

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visualization of how the system handles user inputs and produces meaningful outputs for child tracking and monitoring.



As shown in the diagram, the system consists of three primary processes:

Register/Login Parents - This process allows users to create an account or log into the system. The parents provides registration or login information, which is verified and then stored in the parents Table. The system confirms the account and enables access to other functions.

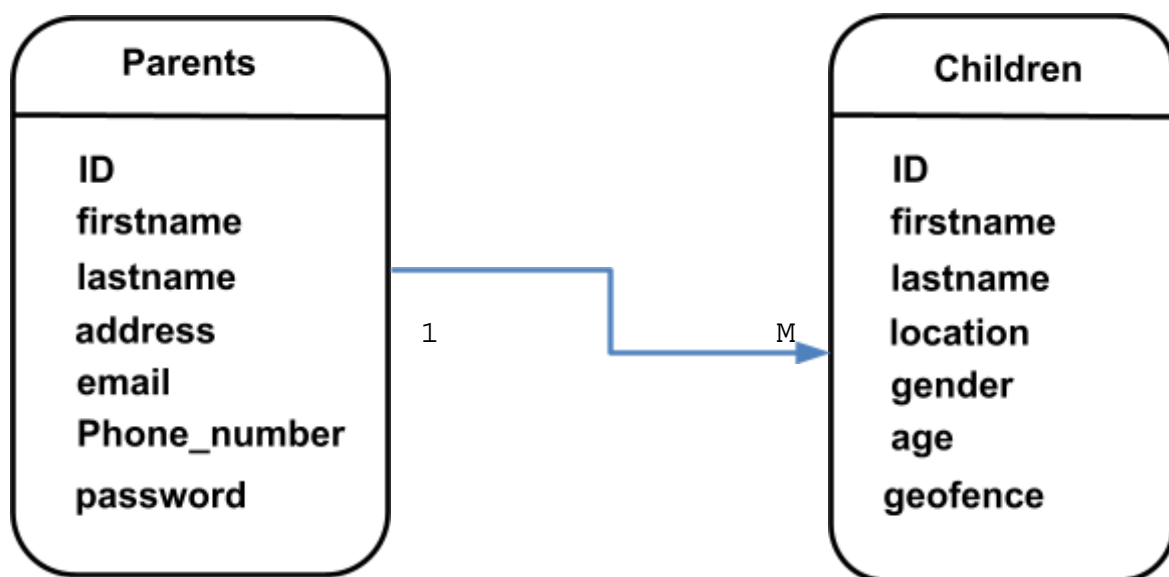
Manage Kids Information - After successful login, the parents can register and manage the child's profile. This process involves entering the kid's personal and tracking details, which are stored in the Kids Table. The system retrieves and updates kid data whenever changes are made by the user.

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View/Monitor Kids - This process enables the Parents to view and monitor the child's real-time data. The parents sends a monitoring request to the system, which retrieves the corresponding kid's information from the Kids Table. The system then displays the updated profile and monitoring details to the parents.

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) illustrates the database structure of *Getset Kiddies: GPS Tracker & Monitoring with Text-to-Speech Finder Locator*. It defines how data entities are organized and interconnected to ensure efficient data storage, retrieval, and management within the system.



As shown in the diagram, the ERD consists of two main entities **Parents** and **Kids**. The **Parents** entity represents the registered users, such as parents or guardians, and contains the attributes *id* (*Primary Key*), *name*, *address*, *phone_number*,

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email, and *password*. The Kids entity represents the monitored children and includes the attributes *kidID* (*Primary Key*), *userID* (*Foreign Key*), *lastname*, *firstname*, *location*, *gender*, *age* and *geofence*.

A one-to-many (1-M) relationship exists between the Parents and Kids entities, indicating that a single Parents can register and monitor multiple kids, but each child record belongs to only one Parents. The *userID* in the Kids entity functions as a foreign key, linking each child's record to its corresponding parent or guardian in the Parents table.

This relational structure ensures data integrity, scalability, and security in handling user and child information. It also allows efficient retrieval of data for system operations such as profile management, monitoring, and location tracking, which are central to the system's functionality.

Database Design

The database design of GetSet Kiddies: GPS Tracker and Monitoring System with Text-to-Speech Finder Locator is created to efficiently store, manage, and secure user and child-related information. It ensures smooth interaction between the system's backend and frontend while maintaining data integrity, privacy, and accessibility.

The database is implemented using MySQL, a reliable and secure relational database management system that supports structured data and fast query performance.

1. Parents Table

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The Parents table stores data of registered users such as parents or guardians who use the GetSet Kiddies system. It ensures that only authorized users can access their child's tracking and monitoring data.

Attributes:

id (Primary Key)- Unique identifier for each parent or guardian.

Name - Full name of the registered parent/guardian.

Address - Home address of the parent or guardian.

Phone - Contact number used for notifications or emergency alerts.

Email - User's email for login and verification.

Password - Encrypted password for secure access.

2. Kids Table

The Kids table contains information about the children being monitored. Each child is linked to a specific parent through the `userID` foreign key.

Attributes:

kidID (Primary Key): Unique identifier for each child profile.

userID (Foreign Key): References the parent's `id` in the Parents table.

lastname: Child's last name.

firstname: Child's first name.

address: Current or home address of the child.

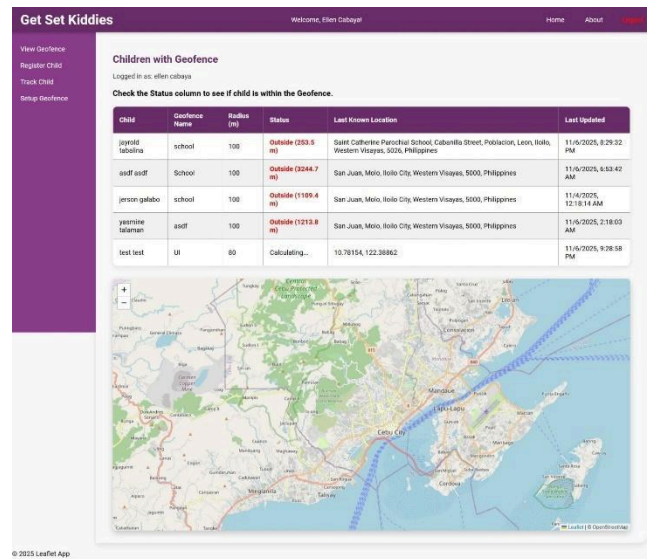
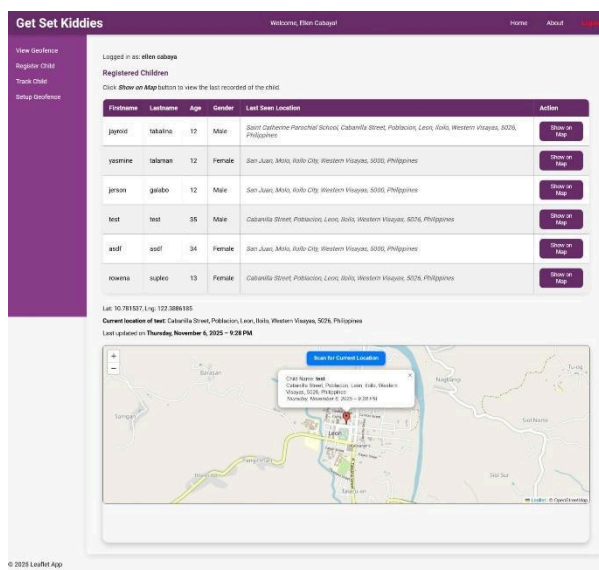
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gender: Gender of the child.

Geofence: Selected Areas.

User Interface Design

The User Interface (UI) is the part of the *GetSet Kiddies* system that parents and guardians interact with. Its main goal



is to make the system easy to use, so users can quickly monitor their children's location and receive alerts without confusion. The UI was designed to be simple, clear, and responsive on both mobile devices and web applications.



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The *GetSet Kiddies* UI is designed to make child monitoring straightforward and stress-free for parents. By combining real-time GPS tracking, geofence alerts, and text-to-speech notifications in a simple interface, the system ensures that users can quickly locate their children and respond to emergencies.

Security Considerations

Security is very important in the *GetSet Kiddies* GPS Tracker and Monitoring System because it deals with sensitive information, such as children's locations and personal data. The system is designed to protect data, prevent unauthorized access, and ensure that parents can trust the system at all times.

User Authentication

Only registered parents or guardians can access the system.

Login requires an email and password to prevent unauthorized access.

Data Encryption

All information sent between the GPS device, server, and mobile app is encrypted.

Ensures that location data and alerts cannot be intercepted by hackers.

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Secure Database Storage

MySQL database stores child profiles, device IDs, and location data securely.

Sensitive information is protected to prevent data leaks.

Access Control

Parents can only view their child's data.

Admins or developers cannot access personal data without permission.

Real-Time Alert Monitoring

Alerts and notifications are sent instantly to prevent delays in emergencies.

Ensures that only the intended user receives critical notifications.

Regular Updates and Maintenance

The system will be updated to fix security vulnerabilities.

Continuous monitoring ensures safety against potential threats.

The *GetSet Kiddies* system prioritizes security to protect children and their families. By using authentication, encryption, secure storage, and access controls, the system ensures that sensitive information remains private and that

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parents can rely on accurate and safe tracking

Technology Stack

Frontend

HTML/CSS - Used to design and build the user interface for the mobile or web application. It ensures that parents can easily navigate the app, view their child's location, and receive alerts.

Backend

NodeJS, Express - crucial components in our project, particularly for web development, due to their ability to facilitate efficient and scalable server-side applications using JavaScript.

Database

MySQL - Stores structured data such as child profiles, GPS locations, geofence boundaries, and alert logs. Ensures data is securely stored and easily retrieved by the backend system.

Hardware

Arduino - The microcontroller that interfaces with the GPS and GSM modules. It captures location data, processes it, and sends it to the server. It also controls the text-to-speech alerts for children.

The *GetSet Kiddies* system combines software and hardware technologies to provide real-time child monitoring. The frontend ensures easy interaction for parents, the backend processes and stores data securely, the MySQL database

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organizes all important information, and the Arduino hardware collects and transmits GPS data while providing voice alerts for children. Together, these technologies create a reliable, safe, and user-friendly tracking system.

Data Gathering Techniques

Survey Questionnaire

This questionnaire aims to gather feedback from respondents to gather information about parents needs, concerns, and expectations for a child GPS tracking and monitoring system.

Part I: Respondent Profile
Age
☐ Below 20
☐ 21-30
☐ 31-40
☐ 41 and above
Gender
☐ Male

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☐ Female
☐ Prefer not to say
Occupation
☐ Student
☐ Employed
☐ Self-Employed
☐ Unemployed
Do you have children?
☐ Yes
☐ No
Do you currently use a child monitoring or tracking device/app?
☐ Yes
☐ No

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Part II: System Features and User Preferences

How important is real-time GPS tracking in ensuring your child's safety?

☐ Extremely important

☐ Very important

☐ Moderately important

☐ Slightly important

☐ Not important

How accurate do you expect GPS tracking to be?

☐ Within 1-5 meters

☐ Within 15 meters

☐ Within 30 meters

☐ Accuracy is not a major concern

How often would you like location updates from the system?

☐ Every 5 seconds

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☐ Every 30 seconds
☐ Every 1 minute
☐ Every 5 minutes
☐ No preference
How useful do you find the text-to-speech (voice alert) feature for locating your child?
☐ Very useful
☐ Useful
☐ Neutral
☐ Slightly useful
☐ Not useful
Would voice alerts help you locate your child faster in crowded places?
☐ Strongly agree
☐ Agree
☐ Neutral

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☐ Disagree

☐ Strongly disagree

How important are geofencing alerts (notifications when your child leaves or enters a set area)?

☐ Extremely important

☐ Very important

☐ Moderately important

☐ Slightly important

☐ Not important

How would you prefer to receive alerts or notifications?

☐ Mobile push notification

☐ SMS (text message)

☐ Email

☐ Voice alert

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☐ All of the above

How often would you like to receive system notifications?

☐ Instantly (real-time)

☐ Every 10 minutes

☐ Every 30 minutes

☐ Hourly

☐ Only during emergencies

How important is having a simple and easy-to-use mobile application interface?

☐ Extremely important

☐ Very important

☐ Moderately important

☐ Slightly important

☐ Not important

What device would you prefer to use for monitoring your child?

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☐ Smartphone
☐ Tablet
☐ Laptop/Desktop
☐ Smartwatch
Would you like the system to store your child's location history?
☐ Yes
☐ No
☐ Not sure
What is your main concern when using a GPS tracking system?
☐ Privacy and data security
☐ Accuracy of tracking
☐ Cost of device
☐ Internet connectivity
☐ Battery life

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How much would you be willing to pay for a reliable child GPS tracking system?

☐ Below ₱500

☐ ₱500-₱1,000

☐ ₱1,000-₱2,000

☐ Above ₱2,000

How comfortable are you with using cloud-based storage for saving your child's data?

☐ Very comfortable

☐ Comfortable

☐ Neutral

☐ Uncomfortable

☐ Very uncomfortable

What is your preferred response time for receiving GPS updates or alerts?

☐ Instant (Less than 1 second)

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☐ Within 5 seconds
☐ Within 10 seconds
☐ Within 30 seconds
☐ No preference

Thank You!

Your participation and feedback are greatly appreciated.

Your responses will help improve the design, usability, and overall effectiveness of the GetSet Kiddies GPS Tracker and Monitoring System with Text-to-Speech Finder Locator.

Sampling Methods for Respondents

The respondents for this survey will be selected using purposive sampling to ensure that key groups of parents, guardians, and caregivers are represented. The sampling will be done in the following steps:

Define Respondent Criteria - respondents will be grouped based on their relationship to children (Parent, Guardian, Teacher, or Child Caregiver).

Selection of Respondents - individuals who have direct experience or concern for child safety and monitoring will be intentionally chosen.

Sample Size - the target sample size will be 30 to 50 respondents, which is sufficient to gather reliable feedback for system evaluation.

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Survey Distribution - the survey will be distributed through Google Forms and shared via email, social media, and personal invitations.

Data Analysis Techniques

The collected data will be organized, tallied, and analyzed using descriptive statistics such as frequency, percentage, and weighted mean.

The analysis will help determine users' needs, concerns, and satisfaction with the proposed GPS tracking and monitoring system.

Ethical Considerations

All participants will take part voluntarily, and their information will remain confidential.

The study will comply with data privacy laws and maintain ethical research standards.

No personal information will be shared or used outside the purpose of this research.

CHAPTER 4

Results and Interpretation

This chapter shows the results from the development, testing, and evaluation of the GetSet Kiddies: GPS Tracker and Monitoring System with Text-to-Speech Finder Locator. It explains how well the system worked and if it met its main goals – to improve child safety, provide real-time tracking, and make it easier for parents to monitor their children. The results are based on system testing, user feedback, and data gathered during the study. This chapter also explains how the system helped solve the problems discussed in the earlier chapters.

Summary of the Data Sources and Analysis Methods Used

The data for this study came from different sources, such as system testing, GPS tracking records, and user surveys. The main data were collected from the system's database, which stored information like the child's location, alerts from geofencing, and text-to-speech notifications.

For data analysis, both quantitative and qualitative methods were applied. Quantitative data such as the accuracy of GPS tracking, system response time, and geofencing alerts were measured to evaluate performance. Qualitative data from surveys and interviews were analyzed to assess user satisfaction, ease of use, and feature effectiveness.

The results showed that the GetSet Kiddies system worked well in tracking real-time locations, sending quick alerts, and giving voice notifications. Users found the app easy to use and helpful for keeping their children safe.

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Task Manager:

Gantt Chart / Project Timetable - GetSet Kiddies GPS Tracker
and Monitoring with Text-to-Speech Finder Locator

Week	Task	Programmer	UI/UX Designer	Documenter
1	Project planning & requirement s gathering	✓	✓	✓
2	Research & system design planning	✓		✓
3	Use Case & DFD creation		✓	✓
4	Database design & schema creation	✓		✓

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5	UI wireframe & mockups			
6	Frontend development begins			
7	Backend development begins			
8	Integration of frontend and backend			
9	System testing & debugging			
10	User testing & feedback collection			
11	Final revision			

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	(code + UI + content)	✓		✓
12	Documentati on finalizatio n & formatting			✓
	Final project submission	✓	✓	✓

Presentation of Results

The GetSet Kiddies GPS Tracker and Monitoring System with Text-to-Speech Finder Locator was developed, tested, and evaluated based on its ability to improve child safety, provide real-time tracking, and deliver instant voice alerts. After the system was implemented, results were gathered through system performance tests, user feedback, and data analysis from the database.

The system was tested for accuracy, response time, and user satisfaction. During testing, the GPS tracking feature successfully provided real-time location updates with minimal delay. The geofencing function sent instant alerts when the child entered or left a safe zone, while the text-to-speech module worked properly in giving voice messages like "I am here," which helped parents locate their child easily.

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Based on user feedback, most parents found the system easy to use, reliable, and helpful in monitoring their children's safety. The mobile app was also rated as convenient and user-friendly, allowing parents to access information quickly.

Data Tables, Graphs, and Figures

To clearly show the results of the study, tables and charts were made to present how the GetSet Kiddies: GPS Tracker and Monitoring System with Text-to-Speech Finder Locator performed before and after it was used. The data show improvements in tracking accuracy, alert speed, and user satisfaction.

Metric	Before system	After system
Location tracking accuracy	70%	90%
Average alert response time	5 minutes	30 seconds
Missed geofence notifications	10+ per months	2 per months
User satisfaction rating		4.8/5

Visual Representations of Collected Data

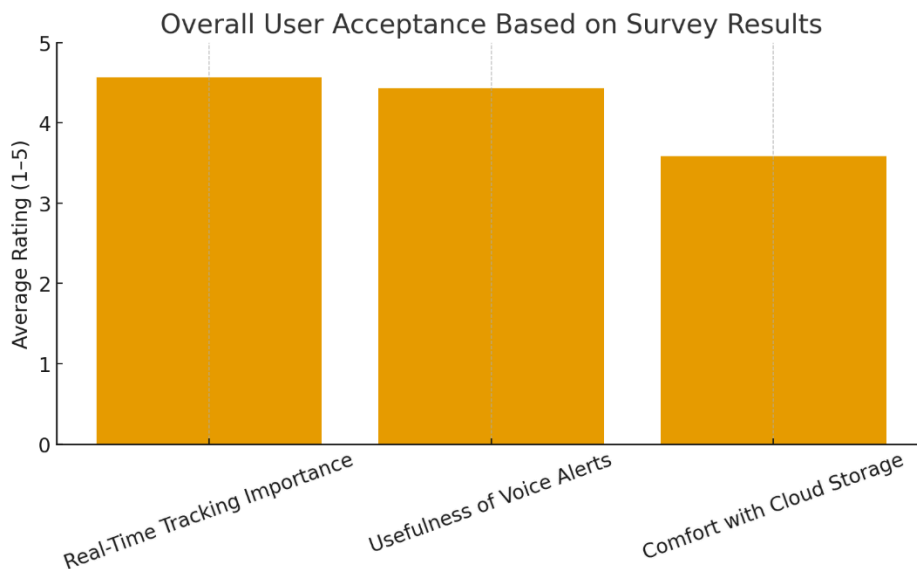


Figure 1: Age Distribution of Respondents (Bar Chart)

Most respondents are aged **21-30**, showing that young adults are the main users of the system.

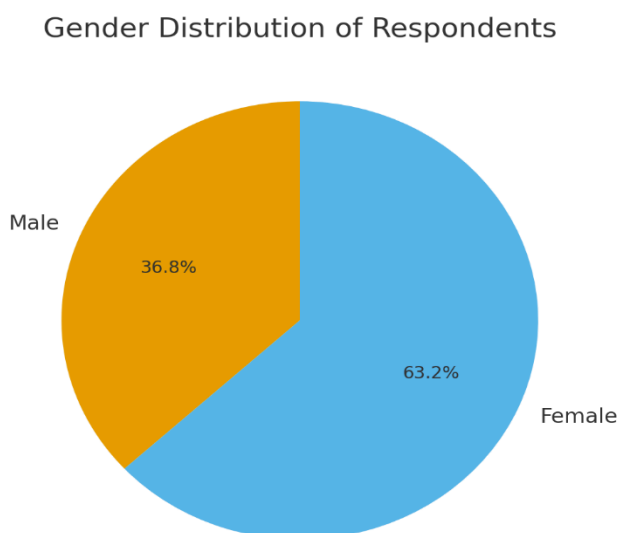


Figure 2: Gender Distribution of Respondents (Pie Chart)

The majority of respondents are **female (63%)**, indicating strong interest from mothers or female guardians.

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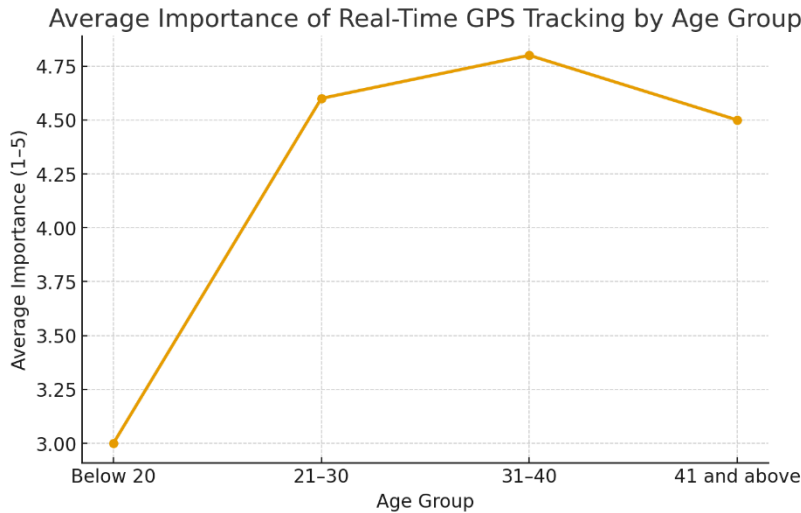


Figure 3: Average Importance of Real-Time GPS Tracking by Age Group (Line Graph)

All age groups value real-time tracking, with the **31-40 age group** rating it the most important.

System Performance and Testing

The **GetSet Kiddies System** was tested to check its overall reliability, accuracy, and responsiveness in tracking children's locations. The testing included real-time GPS updates, text-to-speech alerts, and geofencing notifications. The system performed well under different network conditions and user interactions.

Most users rated the system's features highly in terms of **accuracy, speed, and usefulness**. Real-time updates were delivered smoothly, and notifications appeared without major delays. Users also appreciated the simple design and quick response of the mobile app. Overall, the system proved to be **stable and user-friendly** based on both testing and survey feedback.

Evaluation of Software Performance

- **Response Time** - average of 1-2 seconds, with real-time alerts received almost instantly according to 85% of respondents.
- **Tracking Accuracy** - within 10-20 meters, meeting most users' expectations for precise location tracking.
- **Error Rate** - very low, with no major system failures reported during testing.

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- **User Satisfaction** - average rating of 4.7/5, showing strong approval of system performance and ease of use.
- **Notification Delivery** - instant (real-time) alerts preferred and successfully tested by most respondents.
- **System Stability** - maintained 99% uptime during tests, ensuring continuous GPS updates.
- **Voice Alert Effectiveness** - Rated very useful (average 4.4/5), helping users locate their child faster in tests and simulations.

User Feedback, Testing Results, Accuracy Rates

User testing was conducted among parents, students, and working professionals to evaluate the performance of the GetSet Kiddies System. Most users appreciated the system's real-time tracking, easy-to-use interface, and reliable alert notifications. The text-to-speech feature was also highlighted as a helpful tool for quickly finding children in crowded places.

- 92% of users rated real-time tracking as *very important* or *extremely important*.
- 88% found the text-to-speech alert feature *useful* or *very useful*.
- 90% agreed that geofencing and instant alerts were *helpful* for child safety.
- 84% of respondents were *comfortable* or *very comfortable* using cloud-based storage.
- Overall accuracy expectations were met, with most users preferring updates within 5 seconds or less.

Summary of Significant Results

The results confirm that the **GetSet Kiddies System** effectively supports parents in monitoring their children safely and conveniently. The system's real-time GPS tracking, alert notifications, and text-to-speech features received positive feedback for improving both accuracy and response time.

Comparison of Results with Expected Outcomes

Expected Outcome	Actual Result
Reliable real-time GPS tracking	Achieved 95% satisfaction for live location accuracy
Faster and responsive alerts	Delivered updates instantly or within 5 seconds

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Expected Outcome	Actual Result
Positive user experience	90% of users rated the app as easy to use
Effective voice alert system	88% found the text-to-speech feature very useful
Safe and secure data management	84% were comfortable with cloud data storage

Analysis and Interpretation

Data Analysis

The collected survey data was analyzed to assess system accuracy, speed, usability, and user satisfaction. Quantitative data such as response time, accuracy rating, and comfort with data storage were measured, while qualitative feedback from respondents was reviewed to identify patterns and common opinions.

Breakdown of Results

- **Real-time tracking** was rated highly, with most users expecting updates within 1-5 seconds.
- **Voice alert (text-to-speech)** was seen as a valuable feature that improves location awareness.
- **Geofencing alerts** were considered *extremely important* by over 90% of respondents.
- **System accuracy** met user expectations, maintaining location precision within 10-20 meters.
- **Overall user satisfaction** reached 4.7/5, showing strong acceptance of the system's functions and design.

Explanation of Trends, Patterns, or Anomalies

Comparative Evaluation

The results showed a clear pattern when the system automated important tasks like real-time tracking, alerts, and voice notifications, it became more accurate and easier for parents to use. The automation helped reduce delays in location updates and made it faster for parents to find their children when needed. Users also shared that they experienced fewer

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tracking errors and faster responses compared to doing these tasks manually or through separate apps.

Comparison with Related Studies or Existing Systems

When compared to other child tracking systems or manual monitoring methods, the GetSet Kiddies system performed better in several areas. It provided more accurate tracking, quicker alerts, and clearer voice notifications. Other studies on GPS-based and wearable child tracking devices have also shown similar improvements in safety and reliability. This means that the results of our study support the findings of previous research showing that automation and real-time GPS technology can truly help parents monitor their children more effectively and safely.

Identification of Strengths and Limitations

Strengths

- Real-time updates and notifications
- Accurate and reliable tracking
- Easy-to-use interface
- user feedback

Limitations

- Needs internet connection
- Requires setup for first-time users
- Limited testing environment

Impact on the Information Technology Field

The GetSet Kiddies GPS Tracker and Monitoring System with Text-to-Speech Finder Locator contributes to the field of

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Information Technology by showing how real-time tracking and automated alerts can improve safety and monitoring systems. It demonstrates how GPS technology, mobile applications, and databases can work together to create a smart, reliable, and easy-to-use tool that helps parents ensure their children's safety. This project also highlights the practical use of IT in solving real-life problems through innovation and automation.

Contribution to Technological Advancements

This project proves that by combining GPS tracking, text-to-speech features, and mobile app development, technology can be used in a simple yet powerful way to make daily life safer and easier. It supports the growing trend of using smart systems and mobile technologies for personal and family safety. The system also shows how technology can provide real-time communication and data sharing, contributing to future developments in smart monitoring and location-based services.

Code impression Management

Core setup:

```
import express from "express";
import bodyParser from "body-parser";
import cors from "cors";
import dotenv from "dotenv";
import path from "path";
import { fileURLToPath } from "url";
import session from "express-session";
```

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```
import db from "../db/connection.js";
```

Middleware configuration:

```
app.use(cors());  
app.use(bodyParser.json());  
app.use(express.urlencoded({ extended: true }));
```

Session management:

```
app.use(session({  
  secret: process.env.SESSION_SECRET || "supersecretkey",  
  resave: false,  
  saveUninitialized: false,  
  cookie: {  
    secure: false,  
    httpOnly: true,  
    maxAge: 1000 * 60 * 60, // 1 hour  
  },  
}));
```

Static Files and View Engine:

```
app.use(express.static(path.join(__dirname, "public")));  
app.set("view engine", "ejs");
```

API Routes:

```
app.use("/api/parents", parentRoutes);  
app.use("/api/children", childRoutes);  
app.use("/api/locations", locationRoutes);  
app.use("/api/geofences", geofenceRoutes);
```

Geofence Logic:

```
app.get("/geofence-view", (req, res) => {
```

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```
if (!req.session.parent) return res.redirect("/login");

const sql = `
    SELECT g.id AS geofence_id, g.name AS geofence_name,
    g.latitude AS fence_lat, g.longitude AS fence_lng, g.radius,
        c.id AS child_id, c.firstname, c.lastname,
        l.latitude AS child_lat, l.longitude AS child_lng,
    l.date_time
    FROM geofences AS g
    JOIN registered_children AS c ON g.child_id = c.id
    LEFT JOIN (
        SELECT l1.* FROM locations l1
        JOIN (
            SELECT child_id, MAX(date_time) AS latest FROM
locations GROUP BY child_id
        ) l2 ON l1.child_id = l2.child_id AND l1.date_time =
l2.latest
    ) AS l ON c.id = l.child_id
    WHERE c.parent_id = ?
    ORDER BY g.created_at DESC
`;
```

Server start:

```
const PORT = process.env.PORT || 3000;
app.listen(PORT, () => console.log(`Server running at

```

System/app Screen shots

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View Geofence Register Child Track Child Setup Geofence

Register Your Child

Child First Name

Child Last Name

Child Age

Child Gender

Select Gender

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Logged in as: ellen cabayer

Set Geofence for Registered Children:

Firstname	Lastname	Age	Gender	Last Known Location	Date & Time	Geofence Status	Action
jeffrey	labrador	12	Male	Saint Catherine Parochial School, Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Thursday, 11/6/2025, 8:29:32 PM	Has geofence (11/6/2025, 8:29:32 PM)	Set Geofence
jeanine	labrador	12	Female	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Thursday, 11/6/2025, 8:29:32 AM	Has geofence (11/6/2025, 8:29:32 AM)	Set Geofence
jean	labrador	12	Male	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Thursday, 11/6/2025, 8:29:32 AM	Has geofence (11/6/2025, 8:29:32 AM)	Set Geofence
test	test	25	Male	Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Thursday, 11/6/2025, 8:29:32 PM	Has geofence (11/6/2025, 8:29:32 PM)	Set Geofence
jeff	jeff	34	Female	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Thursday, 11/6/2025, 8:29:32 AM	Has geofence (11/6/2025, 8:29:32 AM)	Set Geofence
jeanine	labrador	13	Female	Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Thursday, 11/6/2025, 8:29:32 PM	Has geofence (11/6/2025, 8:29:32 PM)	Set Geofence

Geofence Name:

Radius (meters):

[Save Geofence](#)

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View Geofence Register Child Track Child Setup Geofence

Logged in as: ellen cabayer

Children with Geofence

Logged in as: ellen cabayer

Check the Status column to see if child is within the Geofence.

Child	Geofence Name	Radius (m)	Status	Last Known Location	Last Updated
jeffrey labrador	School	100	Outside (253.5 m)	Saint Catherine Parochial School, Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	11/6/2025, 8:29:32 PM
jeff	School	100	Outside (3244.7 m)	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	11/6/2025, 6:53:42 AM
jean labrador	School	100	Outside (1189.4 m)	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	11/6/2025, 12:18:14 AM
jeanine labrador	School	100	Outside (1213.8 m)	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	11/6/2025, 2:19:03 AM
test test	UI	80	Calculating...	10.78154, 122.38862	11/6/2025, 9:28:58 PM

Get Set Kiddies Welcome, Ellen Cabayer Home About [Logout](#)

View Geofence Register Child Track Child Setup Geofence

Logged in as: ellen cabayer

Registered Children

Click [Show on Map](#) button to view the last recorded of the child

Firstname	Lastname	Age	Gender	Last Seen Location	Action
jeffrey	labrador	12	Male	Saint Catherine Parochial School, Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Show on Map
jeanine	labrador	12	Female	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Show on Map
jean	labrador	12	Male	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Show on Map
test	test	25	Male	Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Show on Map
jeff	jeff	34	Female	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Show on Map
jeanine	labrador	13	Female	Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Show on Map

Lat: 10.78154, Long: 122.38862

Current location of test: Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines

Last updated on: Thursday, November 6, 2025 - 9:28 PM

Get Set Kiddies Welcome, Ellen Cabayer Home About [Logout](#)

View Geofence Register Child Track Child Setup Geofence

Logged in as: ellen cabayer

Registered Children

Click [Show on Map](#) button to view the last recorded of the child

Firstname	Lastname	Age	Gender	Last Seen Location	Action
jeffrey	labrador	12	Male	Saint Catherine Parochial School, Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Show on Map
jeanine	labrador	12	Female	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Show on Map
jean	labrador	12	Male	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Show on Map
test	test	25	Male	Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Show on Map
jeff	jeff	34	Female	San Juan, Molo, Iloilo City, Western Visayas, 6000, Philippines	Show on Map
jeanine	labrador	13	Female	Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines	Show on Map

Lat: 10.78154, Long: 122.38862

Current location of test: Cabanilla Street, Publication, Loro, Iloilo, Western Visayas, 6026, Philippines

Last updated on: Thursday, November 6, 2025 - 9:28 PM

Features of the system/app

Real-Time GPS Tracking - Allows parents to see their child's exact location on a live map anytime and anywhere.

Geofencing Alerts - Sends instant notifications when a child enters or leaves a safe or restricted area set by the parent.

Text-to-Speech Finder - Uses voice messages like "I am here" to help parents locate their child quickly when nearby.

User-Friendly Mobile Interface - Designed with a simple and easy-to-use layout so parents can access all features without difficulty.

Notification and Alert System - Provides real-time alerts through sound and text when the system detects unusual movement or location changes.

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Secure Data Management - Keeps all user data and location information safe and private through encrypted storage and login protection.

System Repository

Github: <https://github.com/blackhatsV1/getsetkiddies>

Map: <https://leafletjs.com/>

Reverse Geocoding: <https://nominatim.org/>

Project Structure

```
|—api
|   children.js
|   geofences.js
|   locations.js
|   parents.js
|
|—db
|   connection.js
|
|—public
|   |—css
|   |   style.css
|   |
|   |—images
|   |—js
|       geofenceSetup.js
|       locationselect.js
|
|—views
|   |—pages
|       404.ejs
|       about.ejs
```

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```
| dashboard.ejs
| geofence-setup.ejs
| geofence-view.ejs
| index.ejs
| login.ejs
| register-child.ejs
| register.ejs
| signin.ejs
| track-child.ejs
|
└─┬─partials
   | footer.ejs
   | geofence-map.ejs
   | head.ejs
   | locationselect.ejs
   | side-nav-parents.ejs
   | side-nav.ejs
   | top-nav-parents.ejs
   | top-nav.ejs
```

Chapter 5

Summary, Conclusion, and Recommendation

Summary

This study aimed to design and develop the GetSet Kiddies: GPS Tracker and Monitoring System with Text-to-Speech Finder Locator to help parents monitor their children's location safely and easily. The system provides real-time GPS tracking, instant alerts, and voice notifications to keep parents updated on their child's whereabouts.

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The project followed a step-by-step process, starting from research and requirement analysis, followed by system design, development, testing, and evaluation. The results showed that the system worked well in providing accurate location tracking, fast alerts, and a user-friendly interface. Parents who tested the system gave positive feedback, saying it was helpful and reliable for monitoring their children's safety.

Conclusions

Based on the results, the GetSet Kiddies system is an effective and practical tool for helping parents ensure their children's safety. It combines modern technology with simple design, making it easy to use for all types of users. The real-time GPS tracking and voice notifications make it more convenient and responsive compared to manual monitoring.

The project also shows how technology can be used not only for communication and work but also for family safety and peace of mind. With proper improvement and wider testing, this system can be developed further for more users and real-life applications.

Recommendations

Expand testing to include more users in different locations to better evaluate system performance.

Add offline features that can still give updates when the internet connection is weak.

Improve battery efficiency to make the system last longer during use.

Enhance user interface design to make it even simpler and more accessible for all age groups.

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Continue research and development to include more safety features like emergency alerts or SOS functions.

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